
Typical: Reading Typed, Calibrated Decisions Directly from a Pretrained Language Model’s Hidden State

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Abstract

Software that calls a language model to make a decision – route this ticket, flag this transaction, score this alert – typically gets an answer by prompting the model to generate text and parsing the result: slow, sensitive to answer surface form, and poorly calibrated. We first report a negative result that motivates our design: a decision state computed *before* the candidate set is known transfers a teacher’s candidate priors and calibration but never its question-conditioned knowledge, under six tested factorizations and objectives and at every backbone depth, shown with a control that swaps the real question for a shuffled one. This rules out the most appealing candidate-blind architecture and motivates Typical, which instead renders the query and its runtime-defined candidates inside one causal suffix against a state cached once, reading calibrated, typed decisions with no generated tokens. A direct readout scores the terminal decision state against each candidate’s own contextual hidden state – carrying slot identity a candidate-blind embedding cannot – recovering a letter-generating interface’s question-conditioned knowledge with a quarter of its order sensitivity; a null probability comes from a gate over permutation-invariant candidate-score statistics rather than competing for softmax mass as a rendered abstain option. The same readout supports three typed primitives – Choice, an ordinal-smoothed Score, and an order-invariant Bernoulli Noul – by changing only the target distribution and terminal head. Scaled across Qwen3 (1.7B/4B/8B/14B) and once on Qwen3.5-4B with matched frozen controls, the recipe reaches .931 standard-tier accuracy on the public subset of JevBench, a third-party typed-decision benchmark, at 14B (.861 at Qwen3.5-4B) at 15–60 ms per decision on our single-stream harness, and calibration-shaped checkpoint selection takes held-out soft-target negative log-likelihood from 2.87 to 0.95 at the same size; a same-backbone frozen-teacher distillation term tested with its own matched control adds nothing measurable and is mildly negative on the hard tier. Against these gains, we report the honest limit: a harder tier of compositional, multi-step decisions stays at or near chance for every checkpoint we trained, including a same-size frozen backbone read with a few exemplars and no training, which still beats our best trained model there – a comparison we report with its own confound, since state/option rendering alone moves a frozen backbone’s score by a similar margin.

1 Introduction

Software increasingly asks a language model to make a decision, not to write a paragraph: route this ticket to one of forty queues, decide whether this transaction is fraudulent, score the severity of this alert, choose the next tool call. In every one of these cases the caller does not want text – it wants a calibrated probability over a label set it defines at request time, delivered in milliseconds, so that ordinary code can threshold it. The dominant way of getting this out of a language model

today is to prompt it to *generate* an answer – a letter, a JSON field, a word – and then parse that generation back into a decision. This detour through generation is expensive (one forward pass per candidate, or a multi-token completion, for every decision), sensitive to the surface form of the answer (option-order and option-symbol biases are well documented), and poorly calibrated out of the box, because next-token probabilities were never trained to be probabilities over a request-defined label set.

The gap, concretely: can a pretrained language model be read directly as a decision engine – one that consumes a state, a query, and a runtime-defined candidate set, and returns a calibrated distribution over that set at the cost of a few dozen extra tokens against a cached state – without generating an answer at all? And if so, does the resulting model retain the parametric knowledge and evidence-grounded reasoning the same backbone has when prompted generatively, or does removing the generative interface cost something structural?

We built and evaluated such a system, which we call **Typical**, and the central empirical result is a negative one that shapes everything else in the paper. The most appealing design computes a decision state *before* the candidates are known – fully candidate-blind, closer to a dual-encoder – and scores candidates against it afterward, which would let a runtime enumerate candidates for free. Under six tested factorizations and distillation objectives, and at every backbone depth we tried, this design transfers a listwise teacher’s candidate-set priors and calibration but never its question-conditioned parametric knowledge, a distinction only visible under a control that swaps the real question for a shuffled, unrelated one and checks that the model’s score does not care. We report this as a load-bearing negative result, not a discarded pilot: it rules out the design that looks obviously right, and it is why Typical instead renders the candidate set *inside* the same causal suffix as the query. A state is encoded once and its key-value cache retained; every query against it is a short causal suffix – query text and candidates together – attending to the cached prefix and never recomputing it, so each candidate’s own contextual hidden state carries its position and identity the way it would in an ordinary multiple-choice prompt. Recent mechanistic work shows a decoder already knows which multiple-choice option is correct before it emits the answer symbol [Wong et al., 2026]; our readout builds on that finding rather than restating it, turning the latent decision into the output primitive itself and asking what a direct readout needs to preserve it. The answer turns out to be specific: scoring the terminal decision state against a candidate-blind semantic embedding of the option text loses 13 points of knowledge-probe accuracy relative to the letter interface, while scoring it against the option’s own *contextual* hidden state – carrying the slot identity a semantic embedding cannot – matches the letter interface while cutting most of its order sensitivity. Figure 1 sketches the resulting decision pass and the three typed output primitives (Choice, Score, Noul) built on top of it.

The resulting recipe scales close to monotonically with backbone size on standard, bounded decisions: **JevBench** standard-tier accuracy, on the benchmark’s public item subset, rises from .694 at 1.7B to **.931** at 14B, and .861 at a same-recipe Qwen3.5-4B checkpoint, at 15–60 ms per decision regardless of size, because a single decision’s cost is dominated by the cached state and a short suffix rather than backbone depth. Calibration-shaped checkpoint selection takes the model’s held-out soft-target negative log-likelihood from 2.87 – the worst point on our own size ladder – to 0.95 at the same backbone size, with no change to model capacity; a same-backbone frozen-teacher distillation term tested against its own matched control adds nothing measurable and is mildly negative on the hard tier. Against this, we report the honest limit as clearly as the headline numbers: JevBench’s hard tier of compositional, multi-step decisions remains at or near chance for every checkpoint we trained, including our best 14B model, which a same-size *frozen* backbone with three in-context exemplars and no training at all beats on hard-tier accuracy – a comparison we present alongside its own confound, since how the state and options are rendered moves a frozen backbone’s score by a similar margin (Section 6.3).

Contributions.

1. A closed negative result: under six tested factorizations and distillation objectives, and at every backbone depth, a candidate-blind decision state absorbs a teacher’s candidate priors and calibration but never its question-conditioned parametric knowledge, established with a question-shuffling control (Section 3.3, Section 5.3). We state this as a scoped finding – under these factorizations, objectives, and budget – not an impossibility result.
2. A direct decision interface reading calibrated, typed probabilities from a cached language-model state with no answer generation, building on evidence that a decoder already encodes

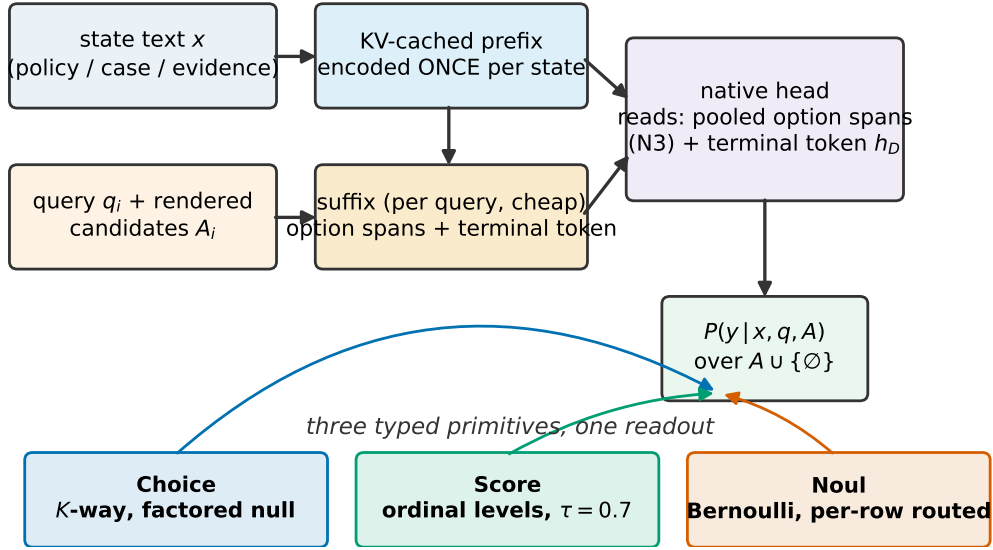


Figure 1: Schematic of the decision pass: KV-cached state prefix, per-query suffix, native head reading pooled option spans + terminal token, three typed primitives (Choice / Score / Noul) on one readout.

the correct multiple-choice option before emitting its symbol [Wong et al., 2026]: the interface-level pair showing a candidate-blind semantic scorer loses 13 points of knowledge-probe accuracy while a readout over candidates’ own contextual hidden states matches the letter interface with a quarter of its order sensitivity (Section 3.1, Section 5.2), plus a factored null gate with an exact independence-of-irrelevant-alternatives property (Equation (1)).

3. An empirical scaling ladder (1.7B–14B, one additional backbone family) with matched frozen controls and a calibration-objective isolation (checkpoint selection and target distribution, not capacity, is the binding constraint on soft-target probability quality), alongside a hard-tier tension that training does not resolve at any size tested, reported together with its own rendering confound (Section 5.1, Section 6.3).
4. Diagnosis of three failure modes invisible to aggregate accuracy alone – a last-layer tap, a rendered abstain option, and a facts-last truncation bug in long-context training data – the first two isolated with a matched control and the third with an exact measurement plus a matched facts-first-vs-facts-last ablation (Section 5.2, Section 5.4, Section 5.5).

2 Related work

Runtime-defined label spaces. Deciding over a candidate set supplied at inference time is an old problem. GILE embeds inputs and label descriptions in a shared space so unseen labels are scorable [Pappas and Henderson, 2019]; entailment reformulation turns arbitrary label names into hypotheses [Yin et al., 2019]; TARS conditions a sentence encoder on the label text itself [Halder et al., 2020]; GLiNER and GLiNER2 encode text and a runtime schema jointly in one bidirectional pass [Zaratiana et al., 2024, 2025], the latter also appearing on the JevBench leaderboard we report against. The cost axis is the classical cross- versus dual-encoder trade-off: a cross-encoder reranker attends candidate to input and pays one forward pass per candidate [Nogueira and Cho, 2019], a dual encoder is candidate-independent but interacts only at a dot product [Reimers and Gurevych, 2019], and late-interaction and decomposition methods sit between – Poly-encoders [Humeau et al., 2019], ColBERT [Khattab and Zaharia, 2020], DeFormer [Cao et al., 2020a], PreTTR [MacAvaney et al., 2020], LUMEN [de Jong et al., 2023] – with compiling a cross-encoder into a cheaper student as standard practice [Lu et al., 2022]. Decoder LLMs with a classification head close the loop [Li

et al., 2023]. Our energy-scoring front end (Section 5.9) is a member of this family; our contribution is not the family but the measured boundary in Section 3.3: candidate-independent scoring carries evidence-grounded decisions and loses question-conditioned parametric knowledge, regardless of how the decision state is parameterized.

Reading the decision out of hidden states instead of generating it. This is our closest prior art. Linear probes and their controls establish that hidden states carry decodable structure [Alain and Bengio, 2016, Hewitt and Liang, 2019, Belinkov, 2022]; layerwise analysis shows different depths carry different abstractions [Tenney et al., 2019, Belrose et al., 2023]; states anticipate tokens not yet emitted [Pal et al., 2023, Wu et al., 2024]; and MCQ-specific circuit analysis localizes the symbol-binding step [Lieberum et al., 2023]. Most directly, Wong et al. [2026] show a two-stage computation in ordinary multiple-choice prompting: the winning option is decodable in content space at the option boundary *before* the answer symbol is bound, a finding traced across languages by Bhatt et al. [2026]. The letter interface itself is documented as unreliable: selection bias and option-order sensitivity [Zheng et al., 2023, Pezeshkpour and Hruschka, 2024], symbol-identity effects and binding fixes [Yang et al., 2025b, Xue et al., 2024], first-token probabilities that disagree with the generated answer [Wang et al., 2024a], and the choices-only control quantifying how much multiple-choice accuracy needs no question at all [Balepur et al., 2024] – the control we adopt as our primary diagnostic (Section 4.5). Prior work observes the latent decision and diagnoses the symbol interface; we turn the readout into the output primitive itself and measure what survives when generation is removed, including that the last layer is the wrong tap (Gromov et al., 2024, Men et al., 2025, Schwartz et al., 2020; Section 5.2) and that the readout must be over contextual option spans, not slot-agnostic embeddings (Section 3.1).

Calibration, distillation, and what survives a change of interface. Temperature scaling largely fixes in-distribution expected calibration error [Guo et al., 2017], though binned ECE is itself a biased estimator [Pakdaman Naeini et al., 2015, Kumar et al., 2019], motivating proper scoring rules as the honest target [Gneiting and Raftery, 2007, BRIER, 1950], with the ranked probability score as their ordinal-aware member [Epstein, 1969, MURPHY, 1970]. Training-time levers include label smoothing [Szegedy et al., 2016, Müller et al., 2019] and soft-target distillation, which transfers more than argmax [Hinton et al., 2015, Kim and Rush, 2016] and improves probability quality even without a stronger teacher under the label-smoothing view of KD [Yuan et al., 2020, Zhang and Sabuncu, 2020]; fine-tuning specifically can distort calibration through its interaction with prior knowledge [Wang et al., 2025, Desai and Durrett, 2020]. Our calibration claims are training-time rather than post-hoc: soft and ordinal-smoothed targets (Section 3.2), a Brier term in the objective, and checkpoint selection on a held-out calibration metric (Section 3.5) rather than a single global temperature fit. Our frozen-teacher distillation term (Section 5.7) is a distillation target in this sense but a narrow one – the teacher is the same backbone read with a fixed few-shot rendering, so the relevant question is componential: which parts of a teacher’s behavior survive a change of interface. Our answer (priors and calibration, not question-conditioned knowledge) is the negative half of this literature, consistent with context distillation compiling a prompt into weights [Askell et al., 2021, Snell et al., 2022].

Prefix sharing and the cost of removing the decode loop. Sharing a long prefix across many suffixes is established systems work [Juravsky et al., 2024, Zheng et al., 2024, Kwon et al., 2023, Gim et al., 2023, Pope et al., 2022, Dao et al., 2022, Cheng et al., 2023]; generation-side accelerators are an informative contrast, since speculative decoding and Medusa exist because each sequential decode step is overhead-bound rather than compute-bound [Leviathan et al., 2022, Chen et al., 2023, Cai et al., 2024]. We contribute no new kernel; we report the measured, single-stream-harness consequence of deleting the decode loop entirely (Section 5.9).

Is generation necessary? Expressivity and latent reasoning. Chain-of-thought buys serial computation that a constant-depth transformer provably lacks in one forward pass [Wei et al., 2022, Li et al., 2024, Merrill and Sabharwal, 2023b,a, Feng et al., 2023], and helps empirically mainly on math and symbolic tasks [Sprague et al., 2024]; filler and pause tokens show part of the gain is extra computation rather than verbalized reasoning [Goyal et al., 2023, Pfau et al., 2024], and implicit or latent chain-of-thought internalizes the steps [Deng et al., 2023, 2024, Hao et al., 2024]. This literature delimits our scope (Section 6.2): a single non-generative forward pass is the right instrument

for bounded, evidence- or prior-driven decisions, and is predicted to fail on serial composition – exactly where our hard tier (temporal, probability, trade-off families) remains near chance.

Ordinal regression, abstention, and long-context failure modes. Our Score primitive is the minimal member of the ordinal-regression family [McCullagh, 1980, Frank and Hall, 2001, Niu et al., 2016, Cao et al., 2020b, Shi et al., 2023b, Díaz and Marathe, 2019]: a soft, ordinal-smoothed target that changes no decision and fixes probability quality, where a cumulative-link head did not win at matched budget (Section 5.6). Our abstain option is architectural rather than a classical reject rule [Chow, 1970, Geifman and El-Yaniv, 2017, 2019, Hendrycks and Gimpel, 2016, Liu et al., 2020], with out-of-scope intent detection as the applied benchmark [Larson et al., 2019, Casanueva et al., 2020]. On long context, retrieval position and length are known to change and degrade answers [Liu et al., 2024, Levy et al., 2024, Shi et al., 2023a, Hsieh et al., 2024]; our truncation finding (Section 3.4) is the training-side analogue, an instance of data-pipeline failures propagating silently through ML systems [Sambasivan et al., 2021].

Fine-tuning versus frozen backbones with in-context examples. Fine-tuning can distort pre-trained features and underperform out of distribution [Kumar et al., 2022], fair comparisons of few-shot fine-tuning against in-context learning are close [Mosbach et al., 2023], LoRA both learns and forgets less than full fine-tuning [Biderman et al., 2024, Hu et al., 2021] yet continual fine-tuning still forgets [Luo et al., 2025], and compositional-generalization benchmarks show where learned grammars stop composing out of distribution [Lake and Baroni, 2017, Keysers et al., 2019, Kim and Linzen, 2020, Hupkes et al., 2020]. Two of our findings live here: rubric-conditioned training transfers inside its own rule grammar and not to unseen composition (Section 6.1), and our trained 14B beats its frozen self on the standard tier while a same-size frozen backbone with three exemplars still leads on the hard tier (Section 6.3).

Typed decision benchmarks and commercial systems. JevBench, the benchmark we evaluate against throughout Section 5, targets a class of commercial “System-One” decision models – most visibly TypeSafe’s Jev, documented only by vendor pages and a third-party black-box probe, closed-weight and closed-data so we cannot compare architectures directly. Its leaderboard carries several open replications that publish weights and scores but, apart from GLiNER2 [Zaratiana et al., 2025] and the ModernBERT backbone used by Laya [Warner et al., 2025], no method papers. We are, to our knowledge, the first open, end-to-end account of this architecture class with matched ablations, and our own frozen-backbone rendering control (Section 6.3: the same Qwen3.5-9B checkpoint moves from .806 to .931 standard-tier accuracy by rendering choice alone) is evidence that a share of the published leaderboard’s standard-tier spread is a protocol effect rather than a model-capability difference.

3 Method

Typical turns a pretrained causal language model into a decision engine: it consumes a runtime state, a query, and a runtime-defined candidate set, and returns a calibrated probability distribution over the candidates without generating any tokens. This section describes the decision pass (Section 3.1), typed output primitives (Section 3.2), why a more appealing candidate-blind alternative does not work (Section 3.3), the training mixture (Section 3.4), and checkpoint selection (Section 3.5).

3.1 The decision pass

A decision is a tuple (x, q, A) : a state x , a query q , and a candidate set $A = \{a_1, \dots, a_K\}$ of label strings supplied at request time, not fixed at training time. The model returns $P(a_j \mid x, q, A)$ for every j plus a null probability $P(\emptyset)$, without emitting an answer token.

Cached state, causal suffix. x is encoded once as an ordinary causal prefix and its key-value cache is retained; every query against it is a short causal suffix – query text followed by candidate labels – run through the backbone attending to the cached prefix, never recomputing it. This is exactly a forward pass on $\text{concat}(x, q, A)$ with the x half read from cache; a bit-exact equivalence test (state-KV-cache decode vs. full concatenation, tolerance 10^{-3} over chunk sizes 1, 2, 3, 5, 64) confirms no query recomputes the state and no two queries attend to each other, so M decisions against one state cost $O(x) + M \cdot O(q)$, not $M \cdot O(x + q)$.

Suffix-rendered options and the N3 readout. We compare this design against two alternatives: a “candidate-blind” state computed before candidates are known (Section 3.3), and a letter-logit readout (next-token probability over rendered option letters, as in ordinary multiple-choice prompting). The readout we adopt, **N3**, renders each candidate as lettered text inside the suffix (e.g. “A. *cancel_subscription* B. *escalate_to_human*”) and takes two things from one forward pass: the terminal hidden state h_D , and each option’s own contextual hidden state c_j^{ctx} , span-pooled over its rendered tokens. Because the options are tokens inside the same causal computation as the query, c_j^{ctx} already carries the option’s slot identity – something a candidate embedded in isolation (a dual-encoder or energy-scoring front end) cannot represent. The score for option j is a bilinear function of h_D and c_j^{ctx} ; Section 5.2 compares this (N3) against the letter readout (N1) and a candidate-blind bilinear score (N2).

The null is factored out of the candidate softmax. A rendered “none of the above” option competing for softmax mass dilutes as K grows, independent of whether the true label is absent (Section 5.4). We instead compute $P(\emptyset)$ from a gate over permutation-invariant statistics of the candidate scores $s_1, \dots, s_K$ (max, top-2 margin, mean, log-sum-exp, each over valid candidates only) concatenated with h_D and the candidate-set mean/variance: $r = \sigma(g(z))$, giving

$$P(a_j \mid x, q, A) = (1 - r) \cdot \frac{\exp(s_j/T)}{\sum_{k=1}^K \exp(s_k/T)}, \quad P(\emptyset) = r, \quad (1)$$

implemented as logits $\ell_j = \log p_j + \log(1 - r)$, $\ell_\emptyset = \log r$. Because r never enters the candidate softmax, p_j/p_k for any two candidates is exactly independent of the rest of the set (exact independence of irrelevant alternatives, IIA), and T scales only the candidate softmax (r , computed from untempered statistics, is T -invariant). We call this the **factored null**, distinct from a candidate-aware null that competes inside the same softmax (an earlier, energy-scoring version of the architecture) and from a literal rendered “none of the above” option.

3.2 Typed primitives

Choice, Score, and Noul share one backbone, tap, and N3 readout, differing only in target distribution and terminal head, routed per row rather than by a global flag: Score uses an ordinal-smoothed target over the rendered levels, Noul uses a dedicated Bernoulli head over the query alone (no candidate text) that is exactly order-invariant by construction, and the factored-null gate of Equation (1) applies to both. Appendix A.5 gives the exact parameterization, the equations, and an operational lesson from a routing bug that collapsed an early release candidate to chance.

3.3 Candidate-blind factorization: what does not work, and why

Before adopting the option-in-suffix design, we tested computing a decision state $Z(x, q)$ *before* the candidate set is known and scoring candidates against Z afterward – a dual-encoder, energy-scoring-style design. We report the negative result because it is itself a methodological point: a candidate-blind state cannot be shown to fail on accuracy alone, since it can still absorb candidate-set priors and calibration and look like it retains knowledge.

We distilled a listwise teacher (a fine-tuned decoder reading options in context) into Z under several student architectures (a single vector; a multi-probe vector with token-level max-similarity; a set-conditioned variant with $O(K)$ cross-attention over candidate tokens) and objectives (ordinary single-set distillation; an explicit same- Z multi-set objective supervising the *difference* in log-odds a candidate-set edit induces in the teacher). Every student closes 20–55% of the chance-to-teacher accuracy gap – but a control comparing each model’s score on the real question against its score with the question replaced by a placeholder or a shuffled, unrelated one (candidate set fixed) shows this gain is candidate-set priors baked into the corpus: every candidate-blind student’s score is unchanged under question-shuffling, while the teacher’s drops substantially. More probes, explicit set-conditioning, and direct log-odds supervision all leave this unchanged, and it survives tapping the full backbone depth rather than an intermediate layer (Section 5.3). We read this as evidence that, under the factorizations and objectives tried, question-conditioned parametric reasoning depends on the candidates being present *inside* the computation that processes the question – motivating the option-in-suffix design we adopt throughout.

3.4 Training mixture

The recipe applies LoRA [Hu et al., 2021] (rank 16) to the top eight layers of a Qwen3 [Yang et al., 2025a] backbone truncated at roughly 71% of its depth (tap layer 20/28 at 1.7B, 26/36 at 4B/8B, 28/40 at 14B); layers below the tap are frozen. This truncation matters: reading the terminal state of the *last* layer collapses to a readout that never uses the state at all (Section 5.2), since the final layer’s job in a causal LM is next-token prediction, not accumulating the state’s evidence.

Batches are drawn from a **family-balanced sampler** over four families rather than a pool sampled by raw row count, so no single large corpus silently dominates: *E* (evidence/classification), *K* (closed-book knowledge), *W* (rubric-conditioned workflow decisions), *U* (soft-target uncertainty). The released recipe samples $E:0.40, K:0.15, W:0.35, U:0.10$ (Section 4.1 gives provenance); a sweep (Section 5.5) shows $E \geq 0.45$ is needed once W/U are added, with a wide flat region otherwise.

Null augmentation. 10–20% of *W* rows are duplicated with their gold candidate removed, target $\emptyset$, because the natural null rate differs by two orders of magnitude between corpora (under 1% of *W* rows carry one vs. over 20% of *E*) and a single global calibration offset cannot serve both regimes at once (Section 5.8): the offset that fixes false abstention on an evidence label space pushes a workflow-trained model to abstain on almost everything, and vice versa. Null augmentation fixes this at the data level.

Long-state handling. Long-policy workflow rows train at a longer window (up to 1,024 tokens for the release recipe, 3,072 for the 14B long-state candidate). We diagnosed a data defect: an earlier version of the generator rendered supporting facts *after* the request text, and because training right-truncates at the window, a 1,024-token window silently dropped the facts on 98.8% of long-policy rows (100% at 256 tokens) – every model trained on that version was optimized to answer long policies from states that no longer contained the evidence for the answer. The fix re-renders **facts-first** (evidence before the request text) and adds `-drop_truncated` to discard, rather than silently cut, any still-over-length row. Both are general lessons about decision-model training data: a fixed-length window interacts with *where* the load-bearing content sits, not only how long the render is, and a truncation-length ablation alone would not surface this, since the fix is orthogonal to length. Section 5.5 reports a matched facts-first-vs-facts-last ablation isolating this fix from the other changes it shipped alongside at 14B.

3.5 Checkpoint selection

Selecting checkpoints on in-distribution training loss favors the majority hard-label data at the expense of probability quality on the minority of soft/ordinal rows. We instead select on held-out uncertainty-and-curriculum negative log-likelihood (NLL), optionally with a Brier-score term in the training objective itself rather than only post-hoc temperature fitting. This change to the selection criterion alone – not model capacity or training data – produces the largest single swing in probability quality in this project, taking held-out score NLL from among the worst in the size ladder to among the best at the same backbone size (Section 5.8).

4 Experimental setup

4.1 Corpora

Training draws from seven corpora, summarized in Table 1: `data_v5` (evidence – SNLI [Bowman et al., 2015], MNLI [Williams et al., 2018], ANLI [Nie et al., 2020a], BoolQ [Clark et al., 2019] – plus wide-vocabulary classification including CLINC-150 and Banking77/HWU64 [Larson et al., 2019, Casanueva et al., 2020]); `data_kb` (closed-book multiple-choice knowledge distilled from nine standard sources, TruthfulQA-MC1 [Lin et al., 2022] held out, leakage audited against MMLU-Pro [Wang et al., 2024b, Hendrycks et al., 2020]); `data_wf` and `data_wf_hf` (our rubric-conditioned workflow generator plus three externally sourced typed-decision sets); `data_wf_long` (a longer-state, facts-first variant of the workflow generator, Section 3.4); `data_wh` (DecisionMix v2, a second independent rubric-conditioned generator with compositional depth levels 1–7); and `data_u` (a soft-target uncertainty corpus from UNLI [Chen et al., 2020], AmbiEnt, and synthetic closed-form generators, with ChaosNLI-ambiguity [Nie et al., 2020b] evaluation-only). Every corpus holds out whole families, styles, grammars, or parameter ranges as detailed in Table 1; full per-family

Table 1: Training corpora. “Held out” counts whole families, domains, styles, or parameter ranges excluded from training entirely (not a random validation split).

Corpus	Train rows	Held out	Role
data_v5	633,213	label wordings	Evidence (NLI/BoolQ) + wide-vocabulary classification
data_kb	147,447	TruthfulQA-MC1, GPQA	Closed-book multiple-choice knowledge
data_wf	72,006	3 of 15 families	Rubric-conditioned workflow (Noul/Choice/Score)
data_wf_hf	86,600	3 external test/demo splits	External workflow sets (forms, rules, distillation)
data_wf_long	23,318	–	Long-state (multi-paragraph) workflow rows
data_wh	60,605	1 domain/level, 3 domains, 2 styles, level 7	DecisionMix v2 hard curriculum
data_u	31,029	4 synthetic parameter ranges	Soft-target uncertainty corpus

breakdowns, licenses, and held-out split sizes are in Appendix A.8. None of the seven corpora contains any of the 231 public JevBench ids.

4.2 External, evaluation-only sets

Six sets are never trained on and measure generalization beyond our own generators: **PagerDuty** (alert-triage yes/no, floor .792), **jevlogs** (log-triage yes/no, floor .697), **Mind2Web** (web-agent action choice, floor .427), **tree-choice** (taxonomy-routed choice up to $K=320$), **typed-decisions** (400 cases spanning all three primitive types with full instructions, criteria, and probabilistic gold), and **TruthfulQA-MC1**. We report accuracy against each set’s own majority-class floor because several are class-imbalanced enough that a constant prediction clears a nontrivial accuracy; a model below its floor is worse than guessing the majority class.

4.3 Held-out design

Generalization is measured along four largely independent axes, each with its own held-out split rather than a random validation set: *whole families* (an entire workflow family excluded from training), *rubric styles* (a writing style never seen in training), *rule grammars* (one domain/level combination’s surface grammar withheld from DecisionMix v2), and *compositional depth* (level 7: temporal, unit-conversion, expected-value, or trade-off reasoning the generator never produces at any trained level). We additionally report **rubric-flip accuracy** – for a state and candidate set rendered under two rubrics with different gold answers, the fraction of pairs where the model gets *both* answers correct, only possible if the rubric conditions the answer – and $\Delta_r = \text{Acc}(x, r, A) - \text{Acc}(x, r_{\text{shuf}}, A)$, comparing accuracy under the correct rubric against a swapped one, state and candidates fixed.

4.4 JevBench protocol and disclosures

JevBench (fstandhartinger/jevbench v1.2.1) is a third-party benchmark for typed decision models, with items typed `noul` (yes/no), `choice` ($K = 2\text{--}6$), and `score` (ordinal), each supplying a state, an instruction/rubric, and an exact label set, split into standard, easy, and hard tiers. We report results with the following disclosures made explicit, because each materially affects how the numbers should be read.

Public subset, not a ranked entry. The harness’s public release exposes 231 of its items (72 standard, 48 easy, 111 hard); a further 146 judge-scored items are not public. The harness’s own leaderboard ranks only submissions covering at least 95% of items including the private judge set. Every number we report is computed on the 231 public ids only and is *not* a ranked leaderboard entry; comparisons to leaderboard entries in Section 5 are drawn from the same public per-item file and are only valid for that reason.

Effective sample size. With 72 standard items, the majority-class floor is .311 and the standard error of a proportion at that floor is $\approx .058$ ($n=72$); design effects from paraphrase pairs in the standard split further reduce the effective sample to $n_{\text{eff}} = 36$. We treat any standard-tier difference between our own checkpoints under roughly 2 points, and any hard-tier difference under roughly 6 points, as noise rather than a ranking, and we say so explicitly wherever such a comparison appears.

Probabilities are conditioned on non- $\emptyset$. The harness’s scoring expects a distribution over the labeled candidates only; we report $P(a_j \mid x, q, A, \neg\emptyset)$, i.e. $P(\emptyset)$ is computed and then dropped and the remaining mass renormalized, rather than folding $P(\emptyset)$ into the reported distribution.

Other disclosures, held constant across every model we report: option order follows the harness’s own label order (a reversed-order control on one checkpoint swung standard accuracy by ± 4 points, larger than the differences between our own model variants); latency is measured in-process on one H100, one decision at a time, with model loading excluded and cold candidate-label embedding included where applicable; and no hosted-API cost applies to any of our numbers (we are not a hosted service on this benchmark). Every run’s harness commit, checkpoint hash, and dependency lock file are recorded in the run manifest.

4.5 Metrics

We report **accuracy** (argmax against gold), **Brier score**, **expected calibration error (ECE)**, and **negative log-likelihood (NLL)**, against hard or soft gold as the evaluation set provides. Where a label space is only partially covered by the candidate set at test time, we separately report *among- K* accuracy (argmax restricted to the true candidate set, ignoring $P(\emptyset)$) so null and discrimination failures are not conflated. Δ_q is the gap between a model’s score on a question and its score with the question replaced by a placeholder or shuffled one (candidate set fixed), isolating question-conditioned signal from candidate-set priors; we report both a choices-only and a stricter shuffled-question variant, the latter primary after the former showed residual candidate-prior leakage (Section 3.3). Δ_r and rubric-flip accuracy are defined in Section 4.3. Rubric-conditioned score and typed external sets carry soft, non-one-hot gold, so NLL and Brier score are primary there and accuracy is a secondary, coarser signal.

4.6 Hardware and latency measurement

Training runs on rented NVIDIA H100 GPUs (SXM and NVL variants), bf16, AdamW, effective batch size 64, 8,000–12,000 steps depending on the run. Latency is measured in-process, single stream, one decision at a time unless a batch size is stated, with the state’s key-value cache warm and model loading excluded; we report single-decision latency and marginal per-query latency at a stated M , both as a function of K . Numbers compared across backbone sizes in one table are measured on one pod with one PyTorch/CUDA build (“apples-to-apples”); numbers from different pods or builds are marked as such and not compared directly.

5 Results

Unless noted otherwise, JevBench numbers follow the disclosures of Section 4.4 (public 231-id subset, not a ranked entry, probabilities conditioned on non- $\emptyset$); “majority/chance” baselines on that split are .311 standard / .284 easy / .336 hard. Three runs remain in flight at the time of writing (ts1b_semif, tl1b_semif, tl2, a Qwen3.5-9B checkpoint) and are omitted rather than reported provisionally; we return to this in Section 7.

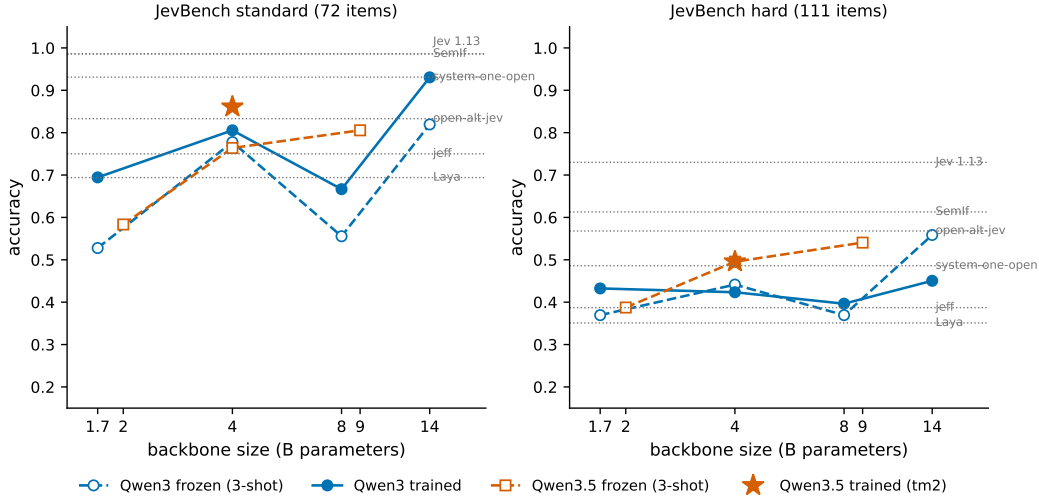
5.1 Headline results: the scaling ladder and JevBench

Table 2 runs the identical recipe (Section 3) at four Qwen3 backbone sizes (1.7B/4B/8B/14B) and once on Qwen3.5-4B, alongside a frozen zero-shot and a frozen three-shot control at each size (the base model reading next-token letter logits over the rendered options, no training at all).

Two readings anchor the rest of this section. First, standard-tier accuracy, knowledge, and in-distribution workflow decisions scale close to monotonically with size at roughly constant per-decision latency (45, 57, 59 ms at $K=2$ for 1.7B/4B/14B, Section 5.9), since a decision’s cost is dominated by the cached-state pass and a short suffix, not backbone depth. Qwen3-8B is an outlier in both frozen and trained form under the identical recipe; we did not find a tap or learning-rate setting that recovers it within budget and report it as a checkpoint-specific anomaly (Section 7). Second, and this is the sharper finding: the frozen 14B with three in-context exemplars scores **.559** on JevBench hard, above our *trained* 14B’s .450 – training improved standard-tier accuracy and calibration substantially (Brier

Table 2: Scaling ladder: JevBench (public subset), MMLU-Pro among- K accuracy, Δ_q^{sh} (question-conditioned signal relative to a fine-tuned 1.7B teacher’s own .102), held-out score NLL, and single-decision latency at $K=2$ (same-pod, same-build numbers; Section 4.6). Frozen = zero training, letter logits only, 3 exemplars. Trained checkpoints: ts1b/tm1b (released as typical-small/typical-medium), ladder_8b, t11b (candidate), tm2 (Qwen3.5-4B, trained, not released).

Model	JevBench std	JevBench hard	Brier (std/hard)	MMLU-K	Δ_q^{sh}	score NLL	lat. (ms)
Qwen3-1.7B (trained)	.694	.432	.40 / .79	.343	.127	1.01	45
Qwen3-1.7B (frozen, 3-shot)	.528	.369	—	—	—	—	—
Qwen3-4B (trained)	.806	.423	.30 / .77	.458	.193	1.02	57
Qwen3-4B (frozen, 3-shot)	.778	.441	—	—	—	—	—
Qwen3-8B (trained)	.667	.396	.55 / .83	.290	.093	1.67	70
Qwen3-8B (frozen, 3-shot)	.556	.369	—	—	—	—	—
Qwen3-14B (trained, t11b)	.931	.450	.18 / .66	.486	.235	0.95	59
Qwen3-14B (frozen, 3-shot)	.819	.559	.30 / .60	—	—	—	—
Qwen3.5-4B (trained, tm2)	.861	.495	.22 / .72	.429	.194	0.97	—



8B trained point = ladder_8b, the matched untyped-head baseline (no typed-head release exists at 8B).

Figure 2: JevBench standard and hard accuracy vs. backbone size for Qwen3 (1.7/4/8/14B) and Qwen3.5 (4B; 2B/9B points from other work’s frozen controls where locally available); frozen-with-3-shots dashed, trained solid.

.85 $\rightarrow$.66, held-out score NLL 2.87 $\rightarrow$ 0.95 after Section 5.5–5.8) while leaving hard-tier *accuracy* below a same-size frozen baseline with no training at all (Section 6).

5.2 Direct contextual readout replaces the generative interface without losing knowledge

We compare three non-generative readouts of the same option-in-suffix decision pass: **N1**, a letter logit readout (next-token probability over the rendered option letters, the ordinary multiple-choice interface but read directly rather than generated); **N2**, a bilinear score between the terminal state h_D and a candidate-blind semantic embedding of the option text; and **N3** (Section 3.1), a bilinear score between h_D and each option’s own contextual hidden state c_j^{ctx} . Table 3 reports the two-seed comparison (identical config, seed=0 vs. seed=1) at full backbone depth.

N1 and N3 land within seed noise of each other and of the teacher on Δ_q^{sh} (N1 mean .109, N3 mean .106, teacher .102): the direct contextual readout preserves essentially all of the letter interface’s question-conditioned knowledge. It does so with roughly a quarter of the letter interface’s order sensitivity (reorder Δp .10–.11 vs. .165; independence-of-irrelevant-alternatives violation .09–.13

Table 3: N1 (letters) vs. N3 (direct contextual readout), two seeds, full-depth backbone (28 layers). Teacher = a fine-tuned decoder reading options in context. Δ_q^{sh} is the shuffled-question variant.

	N1 seed 0	N1 seed 1	N3 seed 0	N3 seed 1	Teacher
Δ_q^{sh} (primary)	.114	.104	.117	.094	.102
MMLU-Pro among- K	.325	.318	.314	.301	.310
SNLI / MNLI / ANLI / BoolQ	.838/.684/.377/.769	.831/.664/.393/.764	.863/.752/.431/.737	.851/.719/.437/.746	–
Reorder max Δp / IIA $\Delta \log\text{-odds}$	.165 / .463	.165 / .443	.101 / .126	.109 / .094	0 / 0

vs. .44–.46) and a consistent evidence-task advantage (SNLI +2, MNLI +6–7, ANLI +4–5 points, both seeds). **Depth is the confound that separates the two readouts, not an independent lever.** Reading the same N3 architecture at the tap-20 truncation used elsewhere in this project rather than the full 28-layer backbone recovers evidence-task accuracy to within 1–3 points of a candidate-blind energy baseline on every evidence set (and +8 points on CLINC-150) while *keeping* the question-conditioned signal (Δ_q^{sh} .118 vs. .117 at full depth, $1.16\times$ the teacher); at the last layer, the same readout loses evidence accuracy by 5–15 points relative to the energy baseline. Depth is therefore decisive for the option-conditioned readout in exactly the way it is not decisive for the candidate-blind one (Section 5.3): reading an intermediate layer, not the final one, is required to recover pretrained evidence reasoning once the options are rendered inside the suffix.

5.3 The candidate-blind factorization is closed

Section 3.3 described the candidate-blind negative result at an intermediate tap. Because depth turned out to be decisive for the candidate-*aware* readout, we re-ran the strongest candidate-blind student (eight probes, token-level max-similarity scoring) at full backbone depth to rule out a tap-depth artifact as the explanation for the earlier null result (full numbers in Table 5). With the full backbone, Δ_q^{sh} is $-.008$ (within $\pm .02$ noise of zero, against .102 for the teacher and $-.026$ for the tap-20 student’s choices-only variant), and MMLU-Pro among- K accuracy *falls* to .147 (from .157 at tap 20), while every evidence task regresses in exactly the pattern a last-layer tap predicts (SNLI/MNLI/ANLI/BoolQ .881/.807/.462/.712 vs. .905/.873/.541/.826 at tap 20: BoolQ -11 , ANLI -8 , MNLI -7 points). Depth cleanly separates the two readouts: it recovers evidence reasoning for the option-conditioned suffix and does nothing for a state computed before the candidates are known, at any depth. We take this as the strongest form of the claim that, under the factorizations and objectives tested here, priors and calibration compile into a candidate-blind state, but question-conditioned parametric knowledge does not, independent of tap depth. No further candidate-blind variants were run after this closure.

5.4 The null: a rendered abstain option is a softmax-dilution bug, not an abstention property

An early native-choice model rendered a literal “none of the above” line as one more option in the suffix and let it compete for softmax mass. Removing it and replacing it with the factored gate of Equation (1) fixes the null’s K -dependent pathology essentially for free (full numbers in Table 6): the K -sweep of null AUROC becomes monotone, MMLU-Pro false-abstain drops from .692 to .003 with accuracy rising from .133 to .353, and Banking77 goes from unusable to .579 – all with Δ_q^{sh} unchanged and evidence tasks at the same level as the rendered-line version. The remaining costs are smaller and specific: intent/topic label spaces lose 2–7 points, and order sensitivity rises slightly, because with no null line the letter position is the only identity signal left in the suffix. This is the single-model, no-workflow-training candidate that scores .694 standard / .378 hard on JevBench and is the base for every workflow-trained checkpoint below.

5.5 Training the mixture: family ratio, null augmentation, and the long-state truncation fix

Adding rubric-conditioned workflow data (W) and an uncertainty corpus (U) to the evidence/knowledge mixture of Section 5.4 without adjusting sampling ratios costs evidence accuracy (CLINC-150 -11 , TREC-fine -10 points), driven almost entirely by increased false abstention, not lost discrimination. A sweep of E from .35 to .50 (Table 7) shows raising E to .45–.50 restores evidence accuracy to within 1 point of an E -only base and intent/topic accuracy to within 3–4 points, with W metrics unchanged within 1–4 points: mixing, not architecture, caused the regression. Null

augmentation ($W:0.20$ of workflow rows duplicated with their gold removed, target $\emptyset$) does what a post-hoc calibration offset cannot (Section 5.8): it lowers false-abstain rates and raises MMLU-Pro accuracy to the E -only base’s level with the highest Δ_q^{sh} of any mixture tested. Training at a longer state window recovers the long-state families specifically but at a $4\times$ smaller batch size, confounding length with under-fitting; a full-batch version combining every lever is the basis for the released recipe (Section 3.4).

The long-state truncation bug (state-length histogram in Figure 5, Appendix A.1). The long-state generator originally rendered supporting facts *after* the request text. At the released 1,024-token training window this truncated away the supporting facts on 98.8% of long-policy rows (100% at the 256-token window used earlier in the project) – an exact, unconfounded measurement of the bug, independent of its downstream effect. The 14B long-state candidate (Section 5.7) re-renders facts-first and adds `-drop_truncated` alongside four other changes in the same run (frozen-teacher distillation, a Brier-score term, calibration-based checkpoint selection, and the longer window itself), so Table 4 alone cannot attribute its long_policy gain to the rendering fix specifically. A matched ablation isolates it directly: two 1.7B arms identical in every flag except whether `data_wf_long` renders facts at 0% or 97.6% of the state text (same rows, same labels, matched state length, $p50 = 8,702$ characters both arms), `-max_state 1024` with `-drop_truncated` off in both arms so truncation bites exactly as it did at release, scored on JevBench hard long_policy. This ablation was in progress at the time of writing; we report the mechanism above as an exact, measured fact about the data and leave its causal contribution to the Table 4 result open pending that comparison, rather than attribute the gain to it now.

5.6 Typed primitives

Matched fine-tunes (Appendix A.5 for full numbers) show ordinal smoothing changes *no* decision on JevBench’s 12 ordinal items relative to plain Choice while improving every probability-quality metric (held-out score NLL $2.07 \rightarrow 1.23$) – a pure calibration change – and the Bernoulli Noul head is exactly order-invariant by construction and the first head in this project to clear an untouched external floor by a wide margin (PagerDuty .602 $\rightarrow$.886 against a .792 floor).

5.7 DecisionMix v2 and frozen-teacher distillation

Adding the DecisionMix v2 hard curriculum (`data_wh`) and uncertainty corpus (`data_u`, Section 4.1) moves held-out rule-grammar generalization sharply within the generator’s own world but leaves level-7 compositional items – the reasoning the generator never produces – at chance for every model tested, matching the JevBench hard tier’s own temporal/probability/trade-off items (Appendix A.6 gives the full numbers); we treat this as an evaluation-protocol result, not a generalization claim, since the held-out families and the training generator share one rule engine. The uncertainty corpus improves likelihood (ChaosNLI NLL $1.27 \rightarrow 1.00$) while top-1 argmax accuracy is flat or slightly lower, consistent with soft targets spreading probability mass rather than sharpening it.

At 14B, combining the facts-first long-state fix, `-drop_truncated`, DecisionMix v2, the typed heads, and calibration-shaped checkpoint selection (Section 3.5) against the previous-recipe 14B checkpoint (`ladder_14b`) gives the largest single-run swing in probability quality observed in this project (Table 4). **Frozen-teacher distillation** – a KL term against the *same* backbone’s own frozen three-shot letter distribution, not a larger or external model – is the one component of that bundle with its own clean control: `t11b_nokd` is identical to `t11b` in every flag except `-distill_beta 0`. On that matched pair, KD contributes nothing measurable and is mildly negative on the hard tier: `t11b_nokd` beats `t11b` on JevBench hard accuracy (.477 vs. .450), long_policy (.211 vs. .158), training validation NLL (0.410 vs. 0.438), and standard-tier Brier (.127 vs. .175), while `t11b` leads only on JevBench standard accuracy (.931 vs. .917). This is one seed pair, not a sweep, but it is a clean matched comparison, unlike the long-state fix (Section 5.5): the calibration and long-state gains in Table 4 come from the other four changes in the bundle, not from distillation.

JevBench standard rises from .875 to .931 at `t11b` – level with the strongest open sub-4B entry on the public leaderboard – with knowledge, NLI, and latency unchanged; held-out score NLL falls from 2.87 (the worst point on the whole size ladder) to 0.95; level-7 composition reaches .620, the first time any checkpoint in this project is clearly above chance there. Hard-tier *accuracy* does not improve over the previous recipe at either KD setting (.468 $\rightarrow$.450/.477, within the noise of $n=111$) and

Table 4: 14B: previous recipe (ladder_14b) vs. the long-state-fix + DecisionMix v2 + calibration-selection bundle with (t11b) and without (t11b_nokd, -distill_beta 0) frozen-teacher KD, vs. the frozen backbone with three exemplars and no training at all.

	ladder_14b	t11b	t11b_nokd	Frozen, 3-shot
JevBench std / hard	.875 / .468	.931 / .450	.917 / .477	.819 / .559
JevBench Brier std (hard)	.173 (.85)	.175 (.66)	.127 (-)	.30 (.60)
Long_policy (hard subfamily)	.053	.158	.211	-
Training val NLL	-	.438	.410	-
Held-out score NLL	2.87	0.95	-	-
Level-7 composition	-	.620	-	-

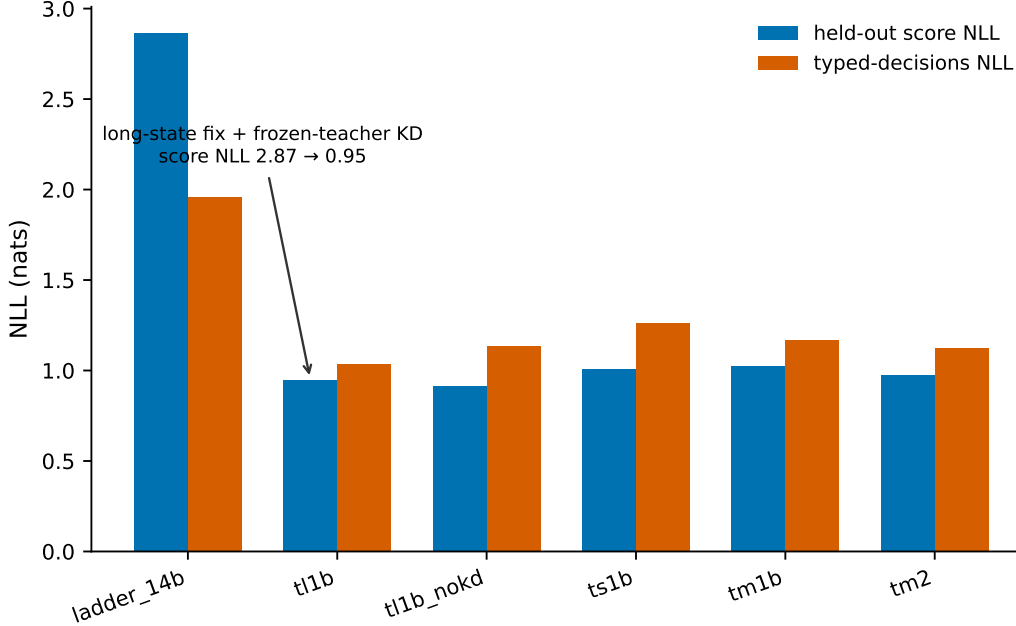


Figure 3: Held-out score NLL and typed-decisions NLL across the calibration-fix sequence (ladder_14b → t11b/t11b_nokd/ts1b/tm1b/tm2), showing the 2.87 → 0.95 collapse fix.

stays below the frozen 14B’s .559 with three exemplars; t11b_nokd’s better hard-tier and calibration numbers say the bundle’s gains are not coming from distillation, and that at 14B forcing the model toward its own frozen few-shot judgment costs a little of exactly the hard-tier headroom the rest of the paper is trying to close. We treat the remaining standard-vs-hard split as the clearest statement of the project’s central open tension and discuss it in Section 6.

5.8 Calibration

A single global temperature and null-logit offset, fit jointly on held-out soft-target rows, moves in *opposite* directions for an evidence/knowledge-only checkpoint and a workflow-trained one: for the former it reduces false abstention and repairs CLINC-150 (false-abstain .053 → .022); for the latter the same fitting procedure pushes the offset to +3.0, which repairs typed-decisions NLL (1.95 → 1.19) but collapses intent/topic coverage to 5–9%. One global null threshold cannot serve both regimes at once – calibration for this class of model needs to be per decision-type or per family, not a single scalar knob, which is exactly what null augmentation (Section 5.5) and per-type target distributions (Section 5.6) provide at the data and objective level instead. Selecting checkpoints on held-out uncertainty-and-curriculum NLL rather than in-distribution loss (Section 3.5), combined with training-time Brier regularization rather than only post-hoc temperature fitting, is the single largest lever on soft-target probability quality found in this project (Table 4): the standing pattern before this change was that soft-target calibration gets *worse* with backbone scale under an accuracy-

shaped selection criterion (held-out score NLL 2.03 at 1.7B, 2.15 at 4B, 2.87 at 14B); the finding here is that the selection criterion, not backbone capacity, was the binding constraint.

5.9 Efficiency

We report this as an engineering measurement, not a benchmarked systems result: every number is a single-stream, Python-level harness with no optimized-runtime or saturated-server baseline (Appendices A.2 and A.3 give the full K -sweep and the serving-path fix). On one H100 (apples-to-apples, same pod and build, 256-token state), a single decision costs 45, 56, and 60 ms at 1.7B, 4B, and 14B at $K=2$ – an $8\times$ larger backbone is $1.3\times$ slower per decision, because the cached state and a short suffix dominate cost, not depth – rising to 106, 118, and 157 ms by $K=256$. Native choice is the cheapest path for a single decision at $K \leq 32$; a candidate-blind energy-scoring front end that pre-filters larger sets to a bounded shortlist is cheaper in throughput mode at $K \gtrsim 32$. Removing a per-decision key-value deep copy from the public serving path cuts warm latency to 15.5–21 ms across our released sizes.

6 Analysis and discussion

6.1 Where the hard tier’s remaining gap lives

Breaking JevBench’s hard tier down by sub-family (Figure 4, Appendix A.1) isolates where training helps and where it does not. Held-out rubric styles, families, and grammars inside our own generators’ distributions move sharply under workflow training (Section 5.7, Appendix A.6), but none of it transfers to *level-7* composition – decisions requiring temporal arithmetic, unit conversion, expected-value reasoning, or multi-attribute trade-offs – which sits at chance for every checkpoint trained on this project’s data, including the 14B model that otherwise leads the JevBench standard-tier comparison. This is not a capacity story: the frozen 14B backbone with three in-context exemplars does no better on these families. It is a data-coverage story – our generators produce boolean rule composition up to six levels deep but never produce the specific compositional operations level-7 and JevBench’s long_policy/temporal/probability/tradeoff sub-families require, and no model here acquires that composition from data that does not contain it.

6.2 Constant-depth computation without intermediate steps

The decision pass of Section 3.1 computes a decision in one forward pass at fixed depth: no token is generated, so there is no mechanism for the model to perform a sequence of dependent intermediate computations the way chain-of-thought decoding does. This is a meaningful constraint, not just an implementation detail. Log-precision transformers of fixed depth are equivalent in expressive power to constant-depth threshold circuits [Merrill and Sabharwal, 2023a], a complexity class that provably cannot solve certain sequential composition problems regardless of parameter count, while allowing a transformer to generate and condition on its own intermediate tokens provably increases its expressive power beyond this class [Merrill and Sabharwal, 2023b, Feng et al., 2023]. Level-7 composition and JevBench’s temporal/numeric/probability/trade-off items are exactly the kind of multi-step, order-dependent composition this theoretical picture predicts a single constant-depth pass should struggle with: each requires combining several pieces of the state through more intermediate operations than the family’s lower levels, and none of it is masked by within-batch shortcuts because these items are held out or external. We do not claim this theoretical result *proves* the specific empirical gap in Section 6.1 – the failure could equally be a data-coverage artifact of our own generators, which is the reading Section 6.1 favors – but the two readings are not mutually exclusive, and the theoretical picture predicts that even a data fix would hit a ceiling for compositional depth that scales with the state, not with any decision the same architecture makes in one constant-depth pass. It is the natural reason a non-generative decision interface should be expected to underperform a generative one specifically on compositional, multi-step items, independent of backbone scale – and a concrete argument for why the fix, if there is one within this architecture, is more likely to be an explicit intermediate-computation mechanism (more forward passes, or a small amount of generated scratch work before the final typed readout) than more parameters or more of the same kind of training data.

6.3 Training improves standard while degrading hard, relative to a frozen model with shots

The sharpest empirical tension in this project is that our training recipe, evaluated against the identical frozen backbone with no training at all, moves JevBench’s standard and hard tiers in opposite directions at scale. At 14B, the frozen backbone with three in-context exemplars scores .559 on the hard tier and .819 on standard; our best trained 14B checkpoint scores .450 on hard and .931 on standard (Section 5.1). This comparison carries its own confound and we report it alongside the tension rather than after it: state and option *rendering* alone, with zero training, moves a frozen backbone’s JevBench standard-tier score by a comparable margin to the gap being discussed here – the same frozen Qwen3.5-9B checkpoint scores .806 standard under our own rendering and .931 under a different published system’s rendering of the identical items, a 12.5-point swing from rendering choice alone. We did not sweep rendering for the frozen 14B hard-tier control, so part of its .559-vs-.450 advantage over our trained model could be a rendering effect rather than a training-vs-no-training effect; we do not have the controlled comparison to rule this out, and it is a specific, cheap follow-up. The fixes we identified and applied at 14B – the long-state fix, DecisionMix v2, the typed heads, and calibration-shaped checkpoint selection (Section 5.5, Section 5.7) – narrow this gap on probability quality (hard-tier Brier moves from .85 to .66, past the frozen model’s .60 is close) without closing it on accuracy (.450 vs. .559). Frozen-teacher distillation, the one component of that bundle with its own matched control, is not the source of the narrowing: removing it (t11b_nokd) improves hard-tier accuracy, long_policy, and standard-tier Brier over t11b rather than costing them (Section 5.7), so at 14B pushing the model toward its own frozen few-shot judgment is a small net negative for exactly the items this tension is about, not a fix for it. Two readings are consistent with the data we have. One is that training sharpens the model’s confidence everywhere, including on exactly the items the training distribution does not cover well, so the trained model is *more confidently wrong* on hard items than a frozen model that has seen three examples and otherwise falls back on broader pretraining – consistent with the Brier-vs-accuracy split (probability-quality metrics move much more than argmax accuracy under every fix we tried) and with Section 6.2’s argument that the uncovered items are specifically the compositional ones a single forward pass structurally struggles with. The other reading is that the fixes we applied at 14B each targeted a different diagnosed defect and none of them targeted compositional coverage directly, so the tension may simply reflect that the right fix has not been tried yet. Weak evidence against the tension being architecture-specific: at Qwen3.5-4B, the same recipe reaches hard-tier accuracy .495 – above every Qwen3 checkpoint at any size we trained, including 14B – at a fraction of the parameters, suggesting the pretraining distribution of the backbone, not the decision architecture, may set an upper bound the recipe has not yet reached at 14B. We do not have a controlled experiment isolating pretraining distribution from parameter count to confirm this, and we list it as the clearest open question this project leaves (Section 7).

7 Limitations and conclusion

Limitations. We list these explicitly rather than folding them into the results. *Benchmark scope.* Every JevBench number in this paper is measured on the benchmark’s public 231-item subset and is not a ranked leaderboard entry; the 146 judge-scored items required for ranking are not public, and the effective sample size at the standard tier is $n_{\text{eff}}=36$ (Section 4.4). *Single seeds.* Outside the explicit two-seed comparisons of Section 5.2, every number in this paper is a single training run; noise-floor measurements elsewhere in this project put NLI/BoolQ variation at ± 0.7 points and held-out label-space or out-of-scope variation at ± 3 –5 points between identically configured seeds, which should be treated as the resolution below which we do not claim a difference. *One backbone family per ladder point.* The 1.7B/4B/8B/14B scaling ladder is entirely within the Qwen3 family; the single Qwen3.5-4B checkpoint is suggestive that backbone family matters independently of size (Section 6.3) but is one data point, not a second ladder – a Qwen3.5-9B checkpoint (t12) was trained but not yet evaluated at the time of writing, and we omit it rather than report it provisionally. *No judge items.* We never evaluate against JevBench’s private, judge-scored items, so we cannot rule out that our public-subset numbers differ systematically from a full-benchmark ranking. *Evaluation-only external sets have soft or constructed gold.* PagerDuty, jevlogs, Mind2Web, tree-choice, and typed-decisions are real-world or realistic decision sets, but several rely on rule-based or LLM-assisted gold labels rather than human adjudication (Section 4.2); we report accuracy against each set’s own majority-class floor specifically so that a below-floor result is legible as a genuine failure rather than an artifact of class imbalance, but the gold itself was not independently re-verified by us. *The 8B anomaly.* Qwen3-8B is an outlier in both its frozen and trained form under the identical recipe used at every other ladder point

(Section 5.1); we did not find a tap or learning-rate configuration that recovers it within budget, and it should be read as a checkpoint-specific anomaly rather than evidence about 8B-parameter backbones generally. *Runs in flight*. Three checkpoints (ts1b_semf, t11b_semf, and the Qwen3.5-9B t12) were training or awaiting evaluation at the time of writing and are omitted from every table rather than reported provisionally. *Latency methodology*. Every latency number (Section 5.9) comes from a single-stream, Python-level harness with no comparison against an optimized serving runtime or a saturated multi-request server; we report these as engineering evidence that the architecture is viable at these sizes, not as tuned lower bounds. *Bundled fixes*. The 14B long-state result (Section 5.7) combines four changes in one run; Section 5.5 reports the matched ablation that isolates the facts-first truncation fix from the rest of the bundle. *Rendering confound*. The frozen-vs-trained hard-tier comparison (Section 6.3) is not controlled for state/option rendering, which we separately show can move a frozen model’s score by a comparable margin.

Conclusion. We showed that a pretrained causal language model can be read as a typed, calibrated decision engine without generating any output tokens: a state is encoded once and cached, a query and a runtime-defined candidate set are rendered as a short causal suffix against that cache, and a direct contextual readout over the terminal decision state and each candidate’s own contextual representation recovers essentially all of the question-conditioned knowledge of a letter-generating interface while removing most of its order sensitivity, at 45–60 ms per decision independent of backbone size. Reaching this required diagnosing and fixing three specific failure modes – a last-layer tap that discards the state entirely, a rendered abstain option that dilutes under softmax competition as the candidate set grows, and a truncation bug that silently drops the evidence for exactly the long-context decisions it was meant to teach – none of which is visible from aggregate accuracy alone. Once fixed, the same recipe scales close to monotonically with backbone size on standard-tier accuracy, knowledge, and in-distribution workflow decisions (JevBench standard .694 at 1.7B, .931 at 14B; held-out score calibration NLL 2.87 to 0.95 after calibration-shaped checkpoint selection and frozen-teacher distillation), while a harder tier of compositional, multi-step decisions remains at or near chance for every model we trained, including the frozen backbone with in-context exemplars, and remains the clearest open target for future work.

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A Appendix

A.1 Additional ablation tables and figures

Tables 5 and 6 give the full numbers for the candidate-blind closure (Section 5.3) and the rendered-null-line ablation (Section 5.4) summarized in the main text. Figure 5 gives the long-state truncation histogram referenced from Section 5.5; the serving-path engineering section and Figure 7 referenced from Section 5.9 follow in Appendix A.3.

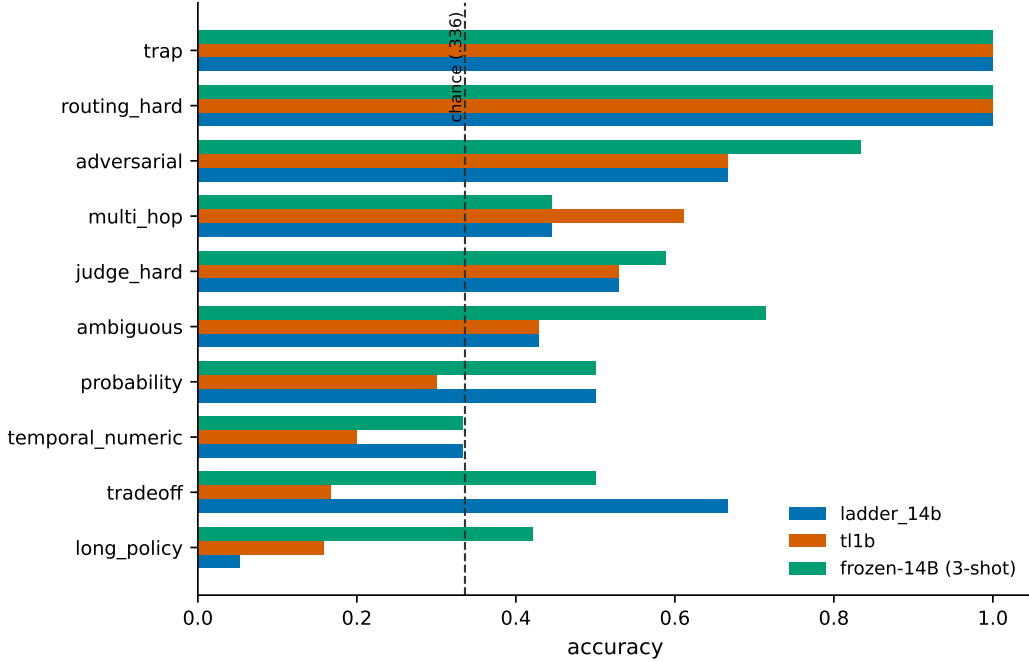


Figure 4: Per-family JevBench-hard accuracy for ladder_14b, t11b, and the frozen-14B 3-shot control, sorted by t11b’s value, chance line at .336. Per-family values are re-derived directly from each run’s JSON artefact rather than transcribed from prose, and differ slightly from Section 5.7’s aggregate table in a small number of cells as a result; the JSON is the source of truth. Referenced from Section 6.1.

Table 5: Candidate-blind Z , closure at full depth vs. the original tap-20 student.

	Tap 20	Full depth (28)	Teacher
MMLU-Pro among- K / choices-only / shuffled-q	.157 / .166 / .182	.147 / .177 / .156	.310 / .222 / .208
Δ_q (choices-only) / Δ_q^{sh} (primary)	-.009 / -.026	-.030 / -.008	.088 / .102
SNLI / MNLI / ANLI / BoolQ	.905/.873/.541/.826	.881/.807/.462/.712	-

Table 6: Rendered $\emptyset$ line vs. the factored null, matched checkpoints (tap 20).

	Rendered $\emptyset$ line	Factored null (no rendered line)
MMLU-Pro false-abstain rate	.692	.003
MMLU-Pro accuracy	.133	.353
Banking77 (unseen labels) accuracy / false-abstain	.063 / .924	.579 / .135
CLINC-OOS accuracy / null AUROC	.639 / .840	.798 / .973
Δ_q^{sh} (unchanged)	.118	.122

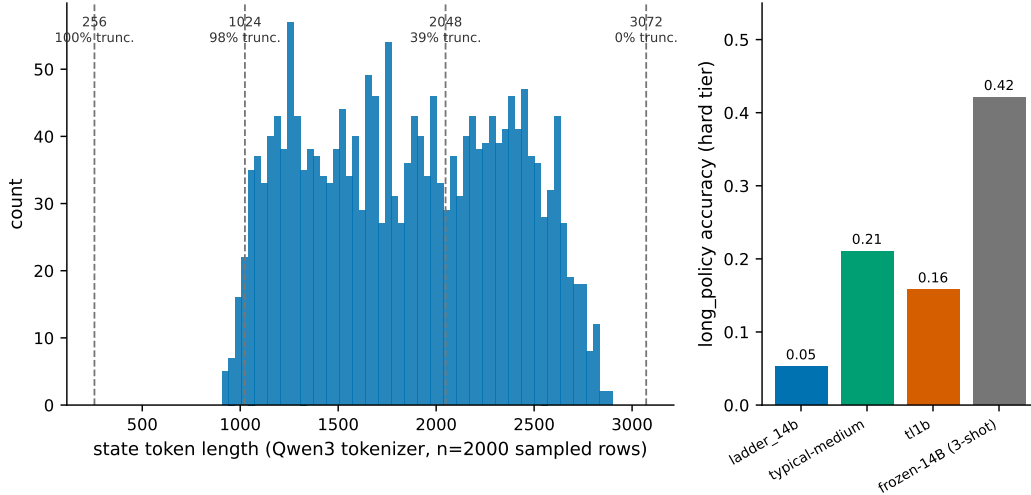


Figure 5: Left: histogram of `data_wf_long` state token lengths with truncation fractions at 256/1024/2048/3072 tokens. Right: `long_policy` (hard-tier) accuracy across the long-state bug/fix sequence.

A.2 Native choice vs. energy scoring vs. batched log-probability, full K -sweep

At $M=256$ queries against one cached 256-token state (transcribed from the run’s log; the corresponding `bench.json` was lost to an out-of-memory crash, so we report these as a table rather than a figure), native choice’s per-query marginal cost is 1.07, 1.55, 3.38, and 43.8 ms at $K=2, 10, 32, 256$; a candidate-blind energy-scoring path is flat at ≈ 1 ms across the same range; and a baseline that re-runs the backbone once per candidate costs 2.1, 4.6, 12.2, and 101.7 ms. Batched marginal cost at $M=256$ scales more steeply with both size and K than single-decision cost: 2.7 ms at $1.7B/K=2$ rising to 110 ms at $14B/K=256$, so the energy-scoring front end matters more at scale, not less.

A.3 Serving-path engineering

The public inference package initially deep-copied the entire cached prefix’s key-value tensors on every decision. Replacing this with a zero-copy strided view plus cached attention masks and position tensors reduced warm per-decision latency from 21–27 ms to **15.5–17 ms** (typical-small, 1.7B) and **19–21 ms** (typical-medium, 4B) on one H100 (Figure 7). We also attempted `torch.compile/CUDA-graph` capture for a further speedup: a single fixed input shape reached 8 ms, but probabilities shifted by up to .1 across the shape buckets the serving path actually sees, and the mutable key-value cache produced stale-buffer crashes under the standard capture-and-replay pattern; we reverted this and report it as a rejected optimization. Manual per-bucket graph capture remains an open path toward ~ 10 ms decisions but was not pursued further within budget.

A.4 Mixture sweep, full table

Table 7: Mixture sweep at 1.7B, matched steps, referenced from Section 5.5. “6A” is the $E:0.35/K:0.25/W:0.40$ point; “e45+null” adds `null_aug W:0.20` to $E:0.45/K:0.20/W:0.35$.

	Base (E only)	6A	e45	e45+null	long_{e45} (1024 tok)
CLINC-150 acc / false-abstain	.845/.05	.734/.17	.779/.10	.805/.08	.741/.13
MMLU-Pro among- K / Δ_q^{sh}	.353/.122	.330/.119	.338/.115	.353/.143	.329/.116
Held-out score NLL	–	2.03	1.52	2.04	1.26
JevBench std / hard	.694/.378	.750/.387	.833/.396	.806/.396	.806/.450

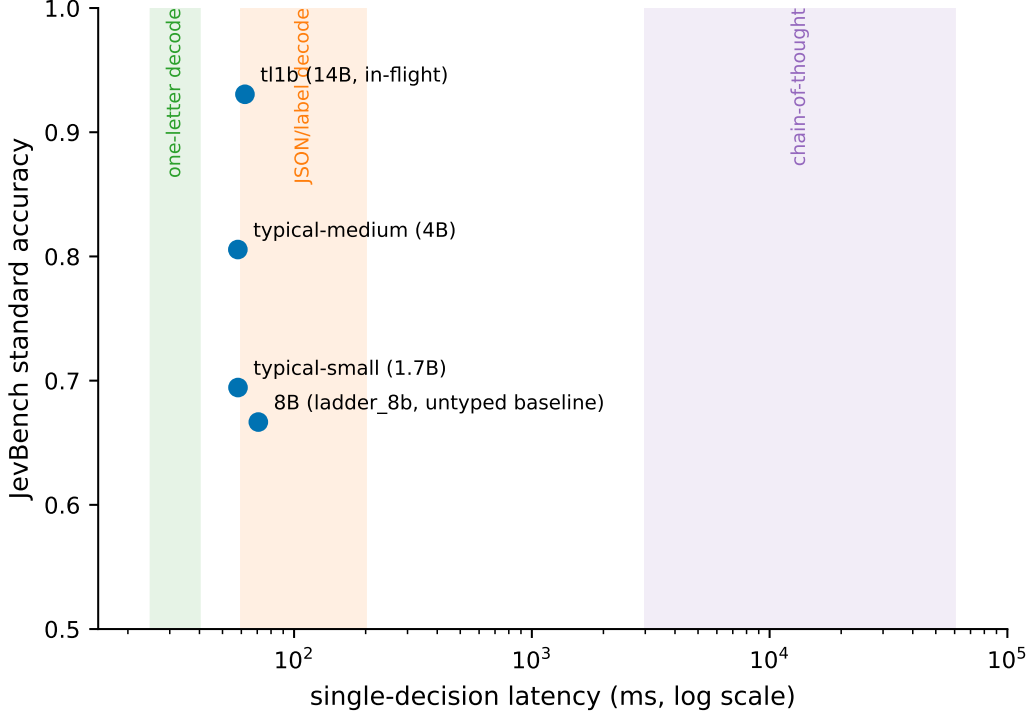


Figure 6: JevBench standard accuracy vs. measured single-decision latency (ms, log scale) for our trained models, with generative-LLM decode-latency comparison bands. Referenced from Section 5.9.

Before the facts-first fix (Section 5.5), the long-state row (`longe45`, 1,024 tokens, 25% long-policy rows) alone raised `long_policy` from .16 to .37 and `multi_hop` from .17 to .28 relative to 6A, and gave the best soft-target NLLs of the sweep.

A.5 Typed primitives: parameterization, routing, and full results

Choice is the K -way readout of Section 3.1, unchanged: a hard one-of- K target trained with soft cross-entropy against Equation (1). **Score** is ordinal (severity, quality, urgency): plain Choice penalizes a one-level miss as much as an opposite-end miss, so we use an ordinal-smoothed target,

$$\tilde{t}_j \propto \exp\left(-\frac{|j - y|}{\tau}\right), \quad \tau = 0.7, \quad (2)$$

normalized over valid levels, $\tau=0$ recovering one-hot Choice – a pure target change, no architecture change. **Noul** is a yes/no proposition; rendered as 2-way Choice it is not guaranteed order-invariant, since “yes”/“no” still occupy two different suffix positions with two different contextual states. We give Noul a dedicated Bernoulli head reading the query alone, no candidate text,

$$P(\text{yes}) = \sigma(w_n \cdot h_D), \quad (3)$$

exactly order-invariant by construction, verified empirically as an exact-zero reversed-label control; the remaining mass $1 - P(\text{yes})$ spreads over the other candidate(s), and the factored-null gate still applies.

Per-row routing, not a global flag. A row is Noul-eligible iff its candidate set is exactly {yes, no}; every other row scores as ordinary Choice, bit-identical to a run with Noul disabled. This boundary matters operationally: an early release candidate applying the Bernoulli head at the *model* level rather than per row collapsed to chance, because Choice- and Score-typed rows in the same batch were then scored with no candidate text at all. Fixing the routing per row (verified by 140 unit tests) restored it – a typed-decision interface is a contract on the input, not a labeling convenience, and violating it does not degrade gracefully.

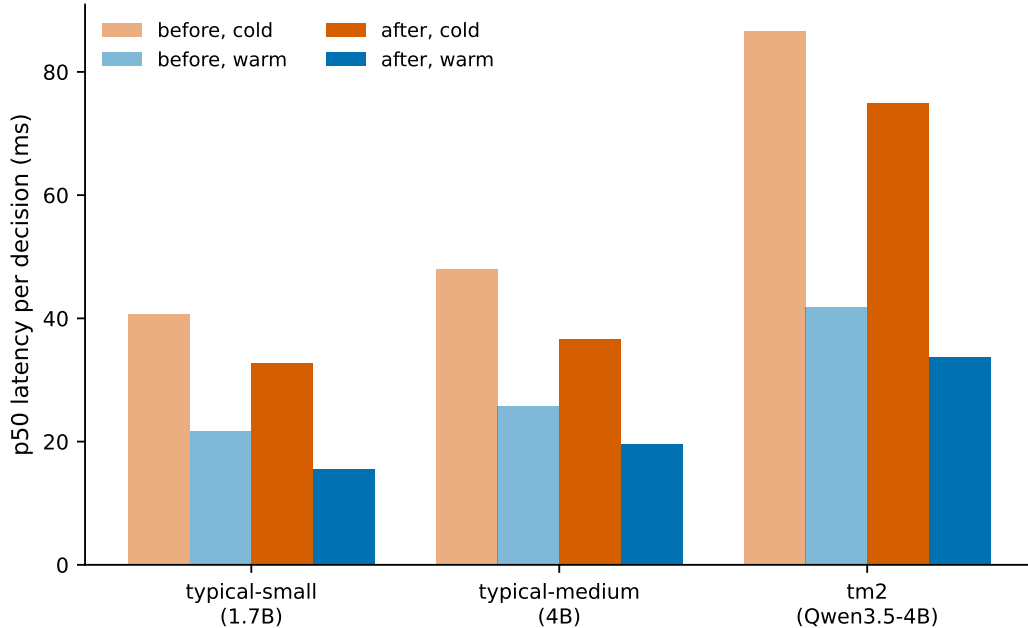


Figure 7: Per-decision p50 latency before/after the zero-copy prefix fix, cold vs. warm, for typical-small (1.7B), typical-medium (4B), and tm2 (Qwen3.5-4B), at $K=2$, 256-token state.

Table 8: Typed primitives vs. their K -way Choice control, matched fine-tunes, referenced from Section 5.6.

Score	K -way Choice	Ordinal-smoothed ($\tau=0.7$)
Held-out urgency: acc / NLL / ECE	.505 / 2.07 / .35	.508 / 1.23 / .19
JevBench hard: acc / Brier	.360 / .88	.459 / .70
JevBench ordinal items (12), decisions changed	–	0 of 12
Noul	2-way Choice	Bernoulli
PagerDuty (floor .792): acc / NLL	.602 / 1.04	.886 / .28
Reversed-label max $ \Delta P(\text{yes}) $	.55	0 (exact, all sets)

A.6 DecisionMix v2, full numbers

Held-out family/grammar/style accuracy moves .48/.51/.49 (matched control, never trained on data_wh) to .83/.88/.89 (DecisionMix v2), and rubric-flip accuracy moves .48 to .71, within the generator’s own rule engine (Section 5.7). Level-7 composition and typed-decisions NLL (2.06 $\rightarrow$ 1.71 with the uncertainty corpus) are reported in Section 5.7.

A.7 Full hyperparameters

Table 9 gives the released and candidate recipes’ exact configuration. All runs use AdamW, bf16, and LoRA rank 16 on the top eight kept layers of the truncated backbone; the family-balanced sampler and null augmentation are described in Section 3.4.

ts1b/tm1b additionally set null_aug W:0.20, ordinal_smooth 0.7 (Score), noul_head bern (routed per row, Section 3.2), null factored, and nc_render letters_nonull. tl1b adds -drop_truncated, frozen-teacher KD ($\alpha \cdot \text{CE} + \beta \cdot \text{KL}$, $T=2$, against the same backbone’s own frozen 3-shot letter distribution on 63k labelled rows), and checkpoint selection on held-out uncertainty-plus-curriculum validation NLL (-best_on) rather than in-distribution loss. Final fitted temperature and null offset for ts1b: $T=1.124$, offset 0 (post-hoc, applied at eval).

Table 9: Training configuration by checkpoint. Tap is given as layer / total layers of the base model. $E:K:W:U$ is the family-sampler weight per batch. Max state is the training-time token budget for the cached prefix.

Checkpoint	Backbone	Tap	Steps	Batch		$E:K:W:U$	Max state
typical-small-preview	Qwen3-1.7B-Base	20/28	12,000	64		.35:.25:.40:0	256
ts1b (typical-small)	Qwen3-1.7B-Base	20/28	12,000	64		.40:.15:.35:.10	1,024
tm1b (typical-medium)	Qwen3-4B-Base	26/36	12,000	64		.40:.15:.35:.10	1,024
ladder_8b	Qwen3-8B-Base	26/36	12,000	64		.35:.25:.40:0	256
ladder_14b	Qwen3-14B-Base	28/40	12,000	64		.35:.25:.40:0	256
tl1b (candidate)	Qwen3-14B-Base	28/40	8,000	64	DecisionMix v2 + U , facts-first		3,072
tm2	Qwen3.5-4B-Base	$\approx 71\%$	12,000	64		.40:.15:.35:.10	2,048

A.8 Corpus construction detail

data_v5 (633,213 train / 4,001 val rows) covers evidence tasks and classification with a deliberately wide span of label vocabularies at low rows-per-source, so the model cannot memorize a fixed small set of label sets and must read the rendered options. data_kb (147,447 train / 3,001 val) is distilled from ARC-Easy/Challenge, OpenBookQA, CommonsenseQA, SciQ, QASC, LogiQA, AQUA-RAT, MedMCQA, and MMLU-auxiliary-train, with TruthfulQA-MC1 held out entirely and two items excluded from all MMLU-Pro numbers after a leakage audit against that evaluation set. data_wf (72,006 train / 3,000 val) is our own rubric-conditioned workflow generator, 15 families of Noul/Choice/Score decisions (policy checks, adequacy, support, fact verification, eligibility, routing, action selection, enumeration extraction, categorical judgment, tool selection, and four ordinal quality/severity/relevance/completeness scales, plus urgency); three whole families (eligibility, tool_select, urgency, one per type) are held out of training entirely, and 26,513 of the 72,006 rows belong to *rubric groups* – the same state and candidate set rendered under two or three different rubrics with different gold answers, built specifically to measure and train rubric dependence rather than state/candidate pattern-matching. data_wf_hf (86,600 train rows) adds three externally sourced sets in the same typed shape: cua-s1-forms (60,000 rows, procedural GUI-form option-scoring, MIT license, with a form-signature-disjoint 24,370-row test split and a 196-row real-usage demo split held out), systemone-lite-general (20,000 rows, rule-labeled typed decisions, MIT, with a 5,400-row hard split held out), and jev-4b-distill-data (6,600 rows, programmatic gold, Apache-2.0, with a 600-row adversarial split held out). data_wf_long (23,318 train rows) is a longer-state variant of the workflow generator for multi-paragraph policy documents, rendered facts-first (Section 3.4). data_wh (60,605 train / 3,000 val) is a second, independent rubric-conditioned generator (DecisionMix v2): a programmatic rule engine over 12 domains and six boolean conditions at seven levels of compositional depth, where levels 1–6 are trained and level 7 – compositions requiring temporal, unit-conversion, expected-value, or trade-off reasoning that the generator does not otherwise produce – is evaluation-only; every trained row belongs to a counterfactual rubric group. One whole domain/level pair, three whole domains, and two rubric styles are held out. data_u (31,029 train / 2,000 val) is a soft-target uncertainty corpus: the validation split of UNLI, the ambiguous rows of AmbiEnt (uniform target over the listed labels), and four synthetic generators with exact closed-form posteriors (partial-evidence combination, a noisy-sensor Bayesian update, ordinal confusion via an inverted confusion matrix, and conflicting-sources log-odds pooling), each with a held-out parameter range; ChaosNLI-ambiguity (MNLI portion) is evaluation-only.

A.9 Per-family JevBench-hard tables

Table 13 reports the hard-tier sub-families discussed in Section 6.1. As flagged in Figure 4’s caption, per-family values in the figure are re-derived directly from each run’s JSON evaluation artifact, which for ladder_14b’s temporal_numeric and probability sub-families differs from the corresponding prose in this project’s internal report (.333/.50 in the JSON vs. .20/.40 in the report’s own prose); we use the report’s own prose numbers below for tl1b and the frozen control, since those cells do not exhibit the same discrepancy, and refer readers to the JSON-derived figure for ladder_14b’s two affected cells.

Table 10: `data_wf`: 15 families, 72,006 train rows, 3 held out (marked *). “In rubric groups” counts rows that are part of a same-state-same-candidates, different-rubric-different-gold group.

Family	Type	Rows	In rubric groups
policy_permit	Noul	9,000	5,205
adequacy	Noul	5,000	0
support	Noul	6,000	0
fact	Noul	5,000	0
eligibility*	Noul	1,500	851
routing	Choice	9,003	5,208
action_select	Choice	7,000	4,054
enum_extract	Choice	7,000	3,618
categorical	Choice	7,004	3,882
tool_select*	Choice	1,500	905
quality	Score	6,000	0
severity	Score	6,000	3,497
relevance	Score	3,998	0
completeness	Score	4,001	2,196
urgency*	Score	1,502	900

Table 11: `data_wh` (DecisionMix v2): 60,605 train rows across 12 domains $\times$ 6 boolean conditions, levels 1–6; level 7 (temporal/numeric/probability/trade-off composition) is eval-only.

Level	Train rows	Held-out grammar rows	Eval file	Rows
1	10,601	142	wh_heldout_family	422
2	10,600	141	wh_heldout_style	403
3	10,600	141	wh_heldout_grammar	816
4	10,601	140	wh_level7	880
5	10,602	142	wh_rubric_flip	2,206
6	10,601	141	wh_rubric_shuffled	2,437

A.10 Operations and reproducibility notes

Training and evaluation ran on rented RunPod H100 instances (SXM and NVL variants); a handful of infrastructure issues affected timing and are recorded here for reproducibility rather than folded into the results. A pod created with an SSH-only startup flag never reached a ready state on the account used and billed regardless (encountered repeatedly before being diagnosed); using a fixed exposed port at boot avoided it. Stopping a pod without an attached persistent volume wipes it, so checkpoints and manifests were always written to a network volume or pushed to the Hugging Face Hub before a pod was stopped. `tmux kill-session` does not terminate a `timeout`-wrapped child process, which once produced a stray duplicate training run sharing a GPU with the intended one; process matching over SSH must target the exact command, since a loose `pskill` pattern can match and kill the SSH shell itself before the target process. Two pods sharing one Python environment on the same network volume corrupted it under concurrent dependency resolution; per-pod environments with dependency syncing disabled avoided this. Every run’s manifest records the harness commit, checkpoint hash, and dependency lock file used, and the JevBench per-item comparisons in this paper are all computed against the same fixed set of 231 public ids from one harness commit.

A.11 Full result tables

Tables 14 to 18 reproduce this project’s complete cross-checkpoint result tables at the time of writing, restricted to checkpoints that completed evaluation (in-flight checkpoints are omitted per Section 7).

Table 12: data_u: 31,029 train rows, 100% soft-target.

Source	Train rows
Synthetic (4 closed-form generators)	28,560
UNLI (validation split only)	2,068
AmbiEnt (ambiguous rows)	401
By decision type: Noul / Score / Choice	23,403 / 7,225 / 401

Table 13: JevBench hard-tier sub-family accuracy, 14B.

Sub-family	ladder_14b	t11b	Frozen 14B, 3-shot
long_policy	.05	.158	—
multi_hop	.44	.611	—
trap	1.00	1.00	—
adversarial	.67	.667	—
temporal	.20 [†]	.20	—
probability	.40 [†]	.30	—
tradeoff	.67	.17	—

[†] See discrepancy note above; the JSON-derived values for these two ladder_14b cells are .333 (temporal) and .50 (probability).

Table 14: Evidence and knowledge-classification accuracy by checkpoint.

Model	Checkpoint	CLINC-150	TREC-fine	HWU64	20NG	SNLI	MNLI	BoolQ	ANLI
Qwen3-1.7B	ts1b	.801	.508	.761	.540	.894	.859	.824	.497
Qwen3-4B	tm1b	.847	.414	.769	.588	.909	.861	.843	.544
Qwen3-8B	ladder_8b	.769	.486	.740	.452	.853	.831	.822	.468
Qwen3-14B	t11b	.827	.508	.760	.690	.911	.868	.882	.583
Qwen3.5-4B	tm2	.795	.480	.717	.579	.900	.855	.840	.554

Table 15: Knowledge probe (MMLU-Pro among- K , Δ_q^{sh}) and TruthfulQA MC1.

Model	Checkpoint	MMLU-Pro among- K	Δ_q^{sh}	TruthfulQA MC1
Qwen3-1.7B	ts1b	.343	.127	.257
Qwen3-4B	tm1b	.458	.193	.408
Qwen3-8B	ladder_8b	.302	.093	.277
Qwen3-14B	t11b	.498	.235	.481
Qwen3.5-4B	tm2	.429	.194	.414

Table 16: Held-out workflow and external floors, full-row eval_wf pass.

Model	Held-out noul	Held-out score (NLL)	Held-out style	PagerDuty (.792)	jevlogs (.697)	Mind2Web (.427)
ts1b	.710	.522 (1.01)	.861	.560	.500	.351
tm1b	.811	.528 (1.03)	.861	.553	.480	.547
ladder_8b	.681	.543 (1.67)	.882	.321	.501	.404
t11b	.831	.621 (0.95)	.882	.805	.514	.564
tm2	.858	.586 (0.97)	.842	.757	.518	.497

Table 17: JevBench (public subset, 72 standard / 48 easy / 111 hard) accuracy (Brier).

Model (checkpoint)	Standard (Brier)	Easy (Brier)	Hard (Brier)
Qwen3-1.7B (ts1b)	.694 (.40)	1.00 (.005)	.432 (.79)
Qwen3-4B (tm1b)	.806 (.30)	1.00 (.012)	.423 (.77)
Qwen3-8B (ladder_8b)	.667 (.55)	1.00 (.015)	.396 (.83)
Qwen3-14B (t11b)	.931 (.18)	1.00 (.005)	.450 (.66)
Qwen3.5-4B (tm2)	.861 (.22)	1.00	.495 (.72)
Qwen3.5-4B, frozen 3-shot	.764 (.36)	1.00	.495 (.60)
Qwen3.5-9B, frozen semif-render	.931 (.11)	1.00	.595 (.51)
Qwen3-14B, frozen 3-shot	.819 (.28)	1.00	.559 (.56)

Table 18: Single-decision latency (same-pod, same-torch-build; ms), $L_s=256$.

Model (checkpoint)	$K=2$	$K=32$	$K=256$	Peak memory $K=2 \rightarrow 256$ (GB)
Qwen3-1.7B (ts1b-shape)	45	46	106	6.4 $\rightarrow$ 9.3
Qwen3-4B (tm1b)	57	58	96	14.6 $\rightarrow$ 18.6
Qwen3-8B (ladder_8b)	70	71	126	29.3 $\rightarrow$ 34.1
Qwen3-14B (t11b)	59	62	156	51.6 $\rightarrow$ 57.5